



CIS 220  
Business Information Systems and  
Analytics

Lecture #23: Data Visualization

November 13<sup>th</sup>, 2025

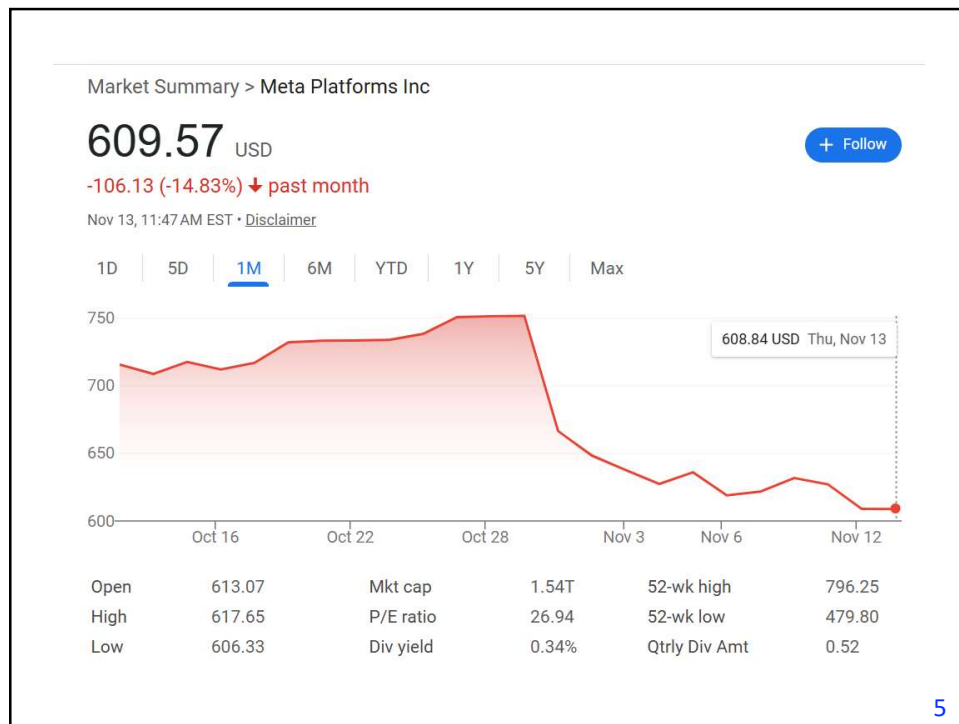
## Administrivia

- Today
  - Tech news
  - Security
  - Finish Information-Age Organizations
  - Introduction to Data Visualization
- Short assignment to watch and make a brief comment on five videos will be due next Tuesday night
  - Will be available sometime Friday
  - Should take under an hour
- Next Tuesday – Start working in Tableau
  - Will send out an email how to get access to Tableau no later than Monday
- Next Thursday – AI Project discussion
  - Will also talk about Exam II, which is on Thursday, December 4<sup>th</sup>
- Questions?

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AI

Meta has an AI product problem

Russell  
Brandom

8:00 AM PST · November 2, 2025

IMAGE CREDITS:  
DAVID PAUL MORRIS/BLOOMBERG / GETTY IMAGES

In the midst of an [unprecedented AI buildout](#), Meta is spending more than most. The company is building two massive data centers, and [reporting indicates](#) there will be as much as \$600 billion in spending on U.S. infrastructure over the next three years.

Those figures might not raise eyebrows in Silicon Valley, but they're starting to make Wall Street nervous.

The issue came to a head this week as [Meta reported quarterly earnings](#), which showed the company's operating expenses jumping \$7 billion year-over-year and nearly \$20 billion in capital expense. It was the result of intense spending on AI talent and infrastructure, which has yet to bring in meaningful [revenue](#) for the company. When analysts [pressed for more specifics](#), Mark Zuckerberg made it clear the spending was just getting started.

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ARTIFICIAL INTELLIGENCE

## Most people worry about AI's impact, but don't think it will come for their jobs, poll finds

New polling from four countries shows that even though workers see AI as a transformative force, they often view their own jobs as safe.



"The AI train appears to be in motion," Weingarten said. "It's leaving the station, but people are not buying a ticket to ride." Weingarten attributed the disparity to a phenomenon called the "awareness action gap."

"People are saying, 'I'm aware that this is coming, but I'm OK for right now. I can spend my spare time pursuing a hobby or thinking about saving money or whatever it is. I don't have to take this action now or today.'"

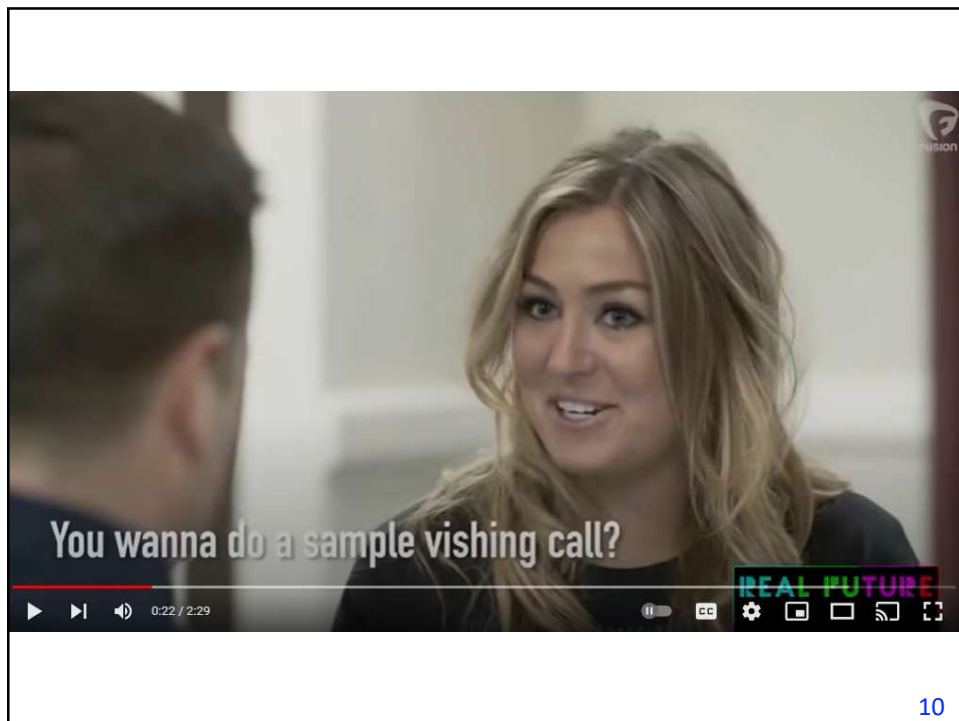
<https://www.nbcnews.com/tech/tech-news/poll-worry-ais-impact-not-jobs-rcna242985>

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## Security Best Practices

- Start with an audit and risk assessment
  - What do we need to protect?
  - What are we afraid might happen?
  - How do we protect it?
- Patch software
- Lock down hardware
  - Standard systems that prevent unapproved software installation
  - Use virtual machines
- Lock down the network
  - Network monitoring is key
  - Firewalls and intrusion detection systems
- Lock down partners
- Lock down systems, e.g., access control and audit trails
  - 2-factor authentication
- Have failure and recovery plans
  - What needs to be done?
  - Close accounts? Issue new accounts?
  - How notify customers and partners?
  - Who works with law enforcement and with the media?
  - Restore from back-up?
  - Can systems be restored without risking more damage?

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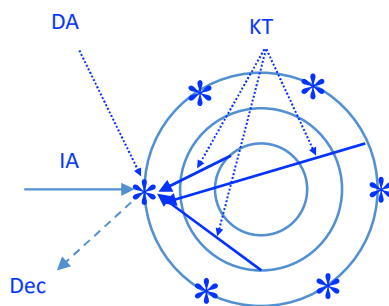


What are the two problems of a traditional hierarchical structure where information is passed up and decisions are passed down?



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## Information-Age Organizations



- Information Awareness (IA)
- Decision Architecture (DA)
- Knowledge Transparency (KT)
- Focus
- Information-Age Network

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## Costs and Benefits of Information-Age Organization

- Costs
- Benefits
- Which is easier to estimate?
- When is an information-age organizational architecture most important?
- Examples?

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## What is the role of IT in each characteristic?

- Information Awareness
- Decision Architecture
- Knowledge Transparency
- Organizational Focus
- Information-Age Network

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Are the characteristics of  
Information-Age Organizations independent?

- Information Awareness
- Decision Architecture
- Knowledge Transparency
- Organizational Focus
  
- Information-Age Network

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What is the impact of AI  
on Information-Age Organizations ?

- Information Awareness
- Decision Architecture
- Knowledge Transparency
- Organizational Focus
  
- Information-Age Network

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## Dell and Hewlett Packard's Personal Computer Businesses

|          | <u>Dell</u> | <u>HP</u> |
|----------|-------------|-----------|
| R&D      | X           | X         |
| Assembly | √           | X         |
| Sales    | √           | X         |
| Service  | √           | X         |

- What does HP do?

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British Petroleum: Why would an oil company need to be an information-age organization?

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## BP Old: Which characteristic are each of these examples a lack of?

- CEO predicted an increase in oil prices, but they remained low
- Some financial proposals required 15 signatures
- Top management learned that a division purchased the rights to search for oil in Qatar from the newspapers
- 12 lines of business in 70 different countries

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## BP New: Which characteristic are each of these an example of?

- Benchmarking against other drillers in Gulf
- CEO joined Board at Intel to learn how to cope with a highly-dynamic environment
- Local-operating units given financial goals, achieve superior performance by discovering their own solutions
- Video-conferencing, job rotation, peer groups
- BPX sold off \$2 billion of assets unlikely to be distinctive
- Outsource utility aspects of IT
- Andrew Oil Field development with partners

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## BP Oil Spill

- What caused spill?
- Lesson
- Which of the above examples may have contributed to a culture that led to spill?
- Lesson
- Example from healthcare
  - DRG-based payments lead to
  - Which leads to
  - So now they are trying to

The AI Race: Meta Debuts Llama 3 | DOJ Scrutiny | Startups to Watch | AI Warfare | AI Unlocks Ancient Secrets

Technology

## Google Consolidates Teams With Aim to Create AI Products Faster

- Research group moved under flagship science unit DeepMind
- CEO alludes to employee protest over AI contract with Israel



Google framed the reorganization as necessary to more sharply focus on launching AI tools and services. Photographer: Patricia de Melo Moreira/AFP/Getty Images



By [Davey Alba](#)

April 18, 2024 at 3:30 PM EDT

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## Google as an Information-Age Organization

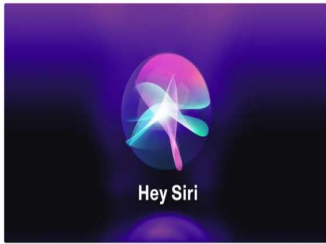
- Information Awareness
- Decision Architecture
- Knowledge Transparency
- Organizational Focus
- Information-Age Network

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Which characteristic would an Apple and Google AI deal enhance? 0

**Apple to Pay Google \$1 Billion Per Year for Siri's Custom Gemini AI Model, Report Says**  
 Apple reportedly chose Google over Anthropic because of cost.

Imed Khan  
 Nov 8, 2023 9:07 am PT 2 min read



Information Awareness 0%

Decision Architecture 0%

Knowledge Transparency 0%

Focus 0%

Information-Age Network 0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at [pollev.com/app](https://pollev.com/app)

Which characteristic does Apple killing their electric car project enhance?

HOME > DIGITAL > NEWS Feb 27, 2024 11:24am PT

**Apple Kills Electric Car Project, Will Shift Many Staffers to Generative AI: Report**  
 By Todd Spangler



Getty Images

Information Awareness 0%

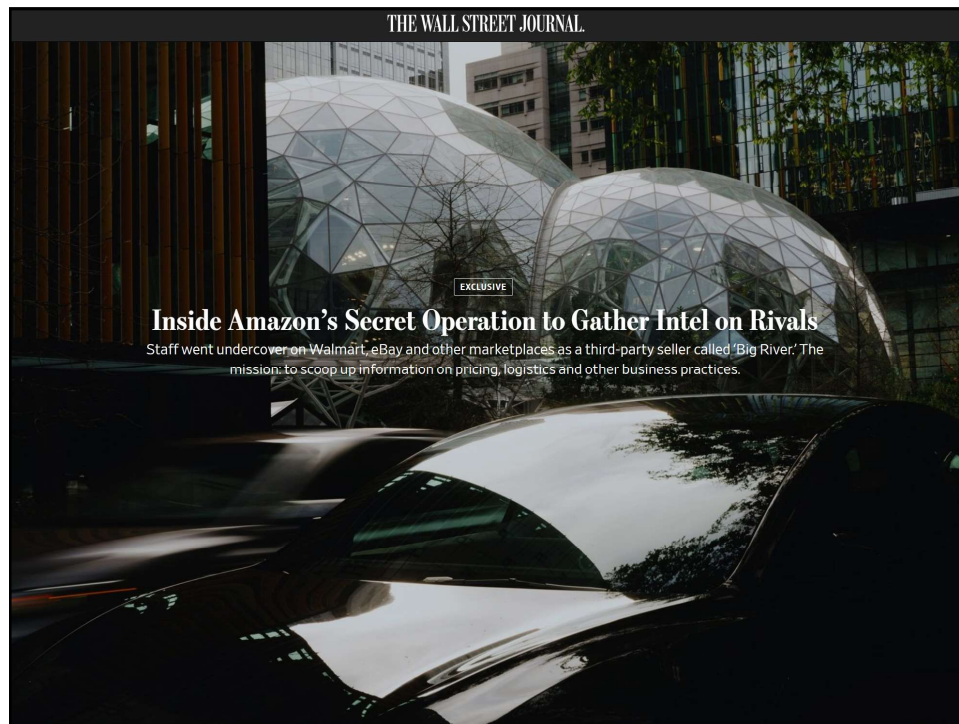
Decision Architecture 0%

Knowledge Transparency 0%

Focus 0%

Information-Age Network 0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at [pollev.com/app](https://pollev.com/app)



By [Dana Mattioli](#) [Follow](#) and [Sarah Nassauer](#) [Follow](#) | Photographs by Grant Hindsley for The Wall Street Journal  
Updated April 17, 2024 9:33 pm ET

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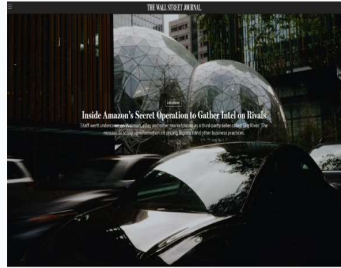
For nearly a decade, workers in a warehouse in Seattle's Denny Triangle neighborhood have shipped boxes of shoes, beach chairs, Marvel T-shirts and other items to online retail customers across the U.S.

The operation, called Big River Services International, sells around \$1 million a year of goods through e-commerce marketplaces including [eBay](#), [Shopify](#), [Walmart](#) and [Amazon](#) [AMZN 0.96% ▲](#) .com under brand names such as Rapid Cascade and Svea Bliss. "We are entrepreneurs, thinkers, marketers and creators," Big River says on its website. "We have a passion for customers and aren't afraid to experiment."

What the website doesn't say is that Big River is an arm of Amazon that surreptitiously gathers intelligence on the tech giant's competitors.

Born out of a 2015 plan code named "Project Curiosity," Big River uses its sales across multiple countries to obtain pricing data, logistics information and other details about rival e-commerce marketplaces, [logistics operations](#) and payments services, according to people familiar with Big River and corporate documents viewed by The Wall Street Journal. The team then shared that information with Amazon to incorporate into decisions about its own business.

### Which characteristic did Amazon's Project Curiosity enhance?



|                         |    |
|-------------------------|----|
| Information Awareness   | 0% |
| Decision Architecture   | 0% |
| Knowledge Transparency  | 0% |
| Focus                   | 0% |
| Information-Age Network | 0% |

Start the presentation to see live content. For screen share software, share the entire screen. Get help at [poller.com/app](https://poller.com/app)

## Why are they firing 600 AI people to save \$150m, while they hire 1 for \$1.5b?

### **Meta Cuts 600 Jobs at A.I. Superintelligence Labs**

The layoffs do not affect Meta's newest AI hires, who are in some cases being paid up to hundreds of millions of dollars. The cuts were focused on correcting an earlier hiring spree.

[Listen to this article](#) (4:05 min) [Show full article](#) [Share](#) [Like](#)



Mark Zuckerberg, Meta's chief executive, at the company campus in Menlo Park, Calif., last month. Meta has hired aggressively to build A.I. in recent years. [Source: Reuters for The New York Times](#)

By Mike Isaac and Ed Tan  
Reporting from San Francisco  
Oct. 20, 2023

## Why are they firing 600 AI people to save \$150m, while they hire 1 for \$1.5b?

- Mark Zuckerberg, Meta's chief executive, has been on a hiring spree to stack his company with top A.I. researchers, including a new chief A.I. officer, Alexandr Wang, earlier this year. The cuts on Wednesday did not affect these newest hires, who have been empowered to develop "superintelligence," or artificial intelligence that exceeds the human brain.
- Instead, the job cuts were aimed at cleaning up the organizational bloat that resulted from three years of building up Meta's A.I. efforts too quickly, two people with knowledge of the matter said. **The layoffs aimed to help Meta develop A.I. products more rapidly,** they said.
- **"By reducing the size of our team, fewer conversations will be required to make a decision, and each person will be more load-bearing and have more scope and impact,"** Mr. Wang wrote in the memo circulated to employees.

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## Information-Age Organizations Wrap-up

- Characteristics of an information-age organization help firms process information more effectively, i.e., make faster and better decisions
- Benefits larger in faster-paced industries
- Questions?

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## Evaluating Visualizations

- Where is the data from?
- What is the visualization showing?
- What's the message?
- In what ways is it a good visualization?
- Anything you don't like about it?
- How might it be misleading?

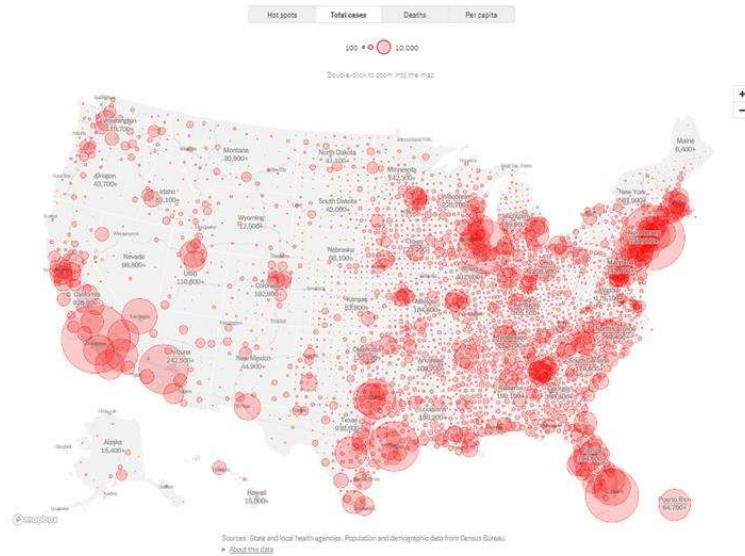
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## Cases in World



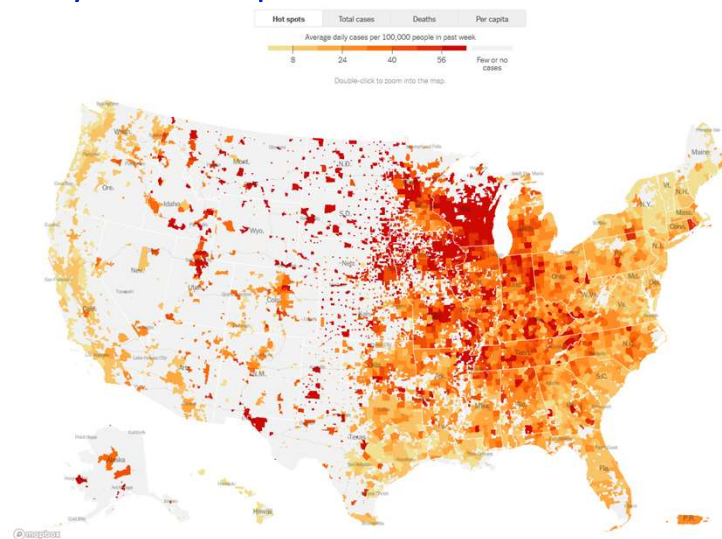
34

## Cases in US



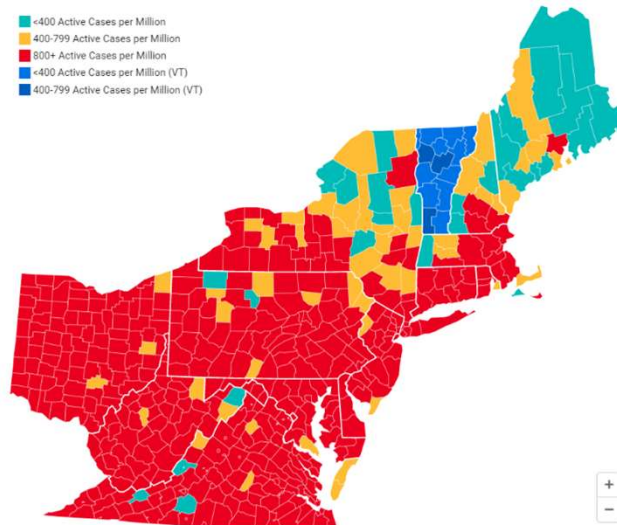
35

## Daily Cases per 100k in last week



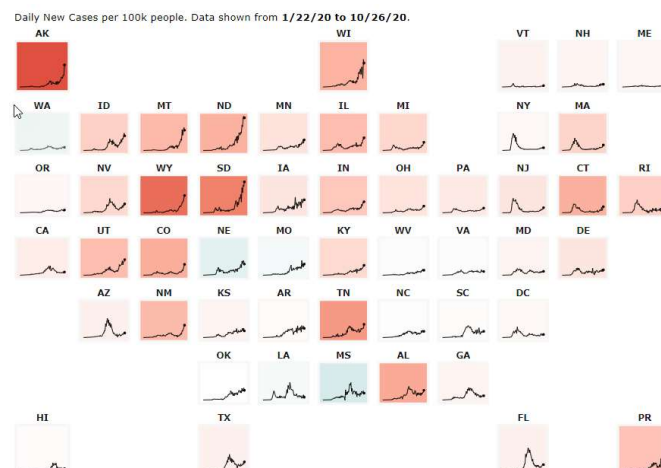
36

## Vermont



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What is the message?  
What is the color?

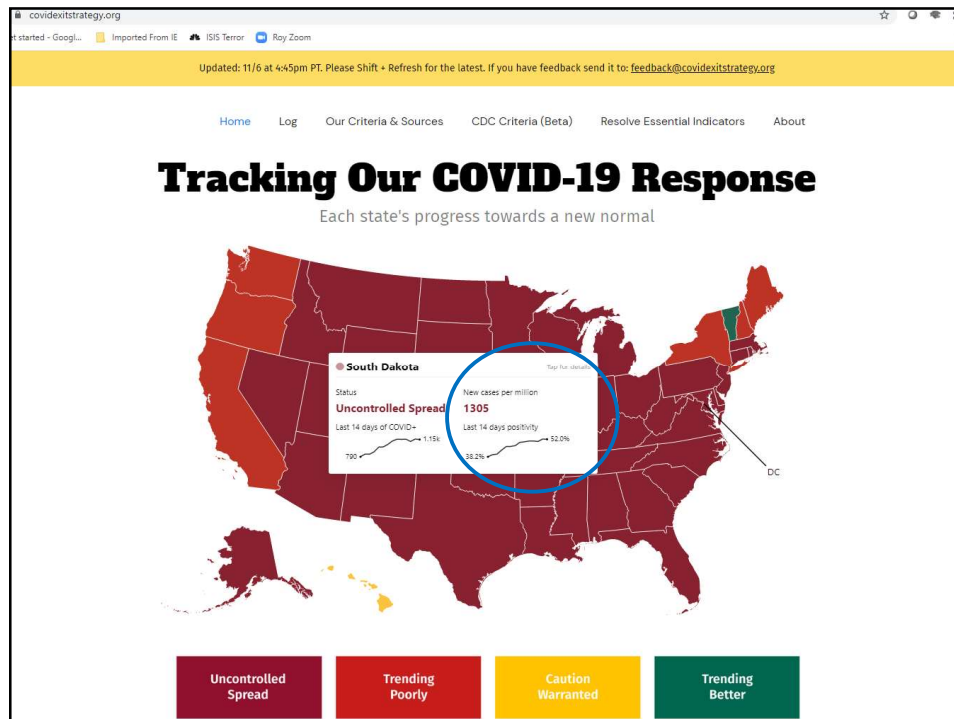


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## COVID-19 Metrics for Trip Planning before vaccines were available

- What metric do you think was most important?
- What does it tell you?
- Can it be gamed?
- Solution
- Note that metrics depend on question. Public health officials may be interested in something else, e.g., hospital bed or ICU bed capacity.

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## Massachusetts Travel Form

### Lower-risk States

Travelers from COVID-19 lower-risk States are not required to fill out the [Massachusetts Travel Form](#) and do not need to quarantine. The Department of Public Health metric for determining lower-risk states for the purposes of Massachusetts' interstate travel policy considers data over two weeks before moving a state from lower risk to high risk. One week of data is the standard for moving states into the lower risk category. Additionally, the threshold of daily cases per 100,000 residents is 10 (which ensures that Massachusetts' standard is in line with other states). States are included on the "lower-risk" list based on meeting two criteria: average daily cases per 100K below 10 AND positive test rate below 5%, both measured as a 7-day rolling average.

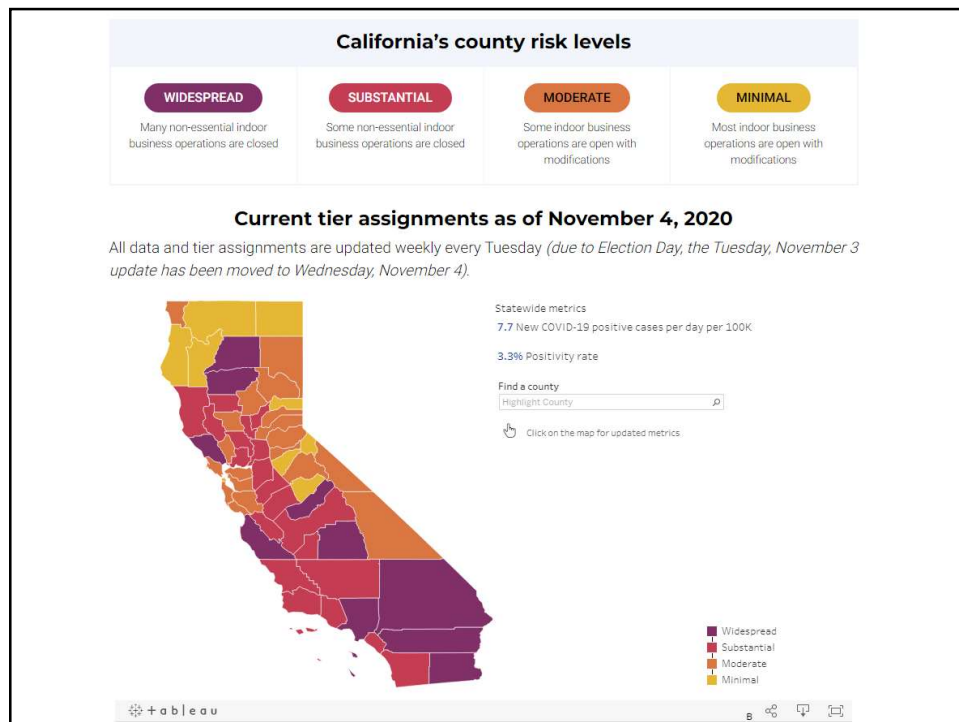
Data is from [covidexitstrategy.org](https://covidexitstrategy.org) as of November 4, 2020.

The current list of COVID-19 lower-risk states includes:

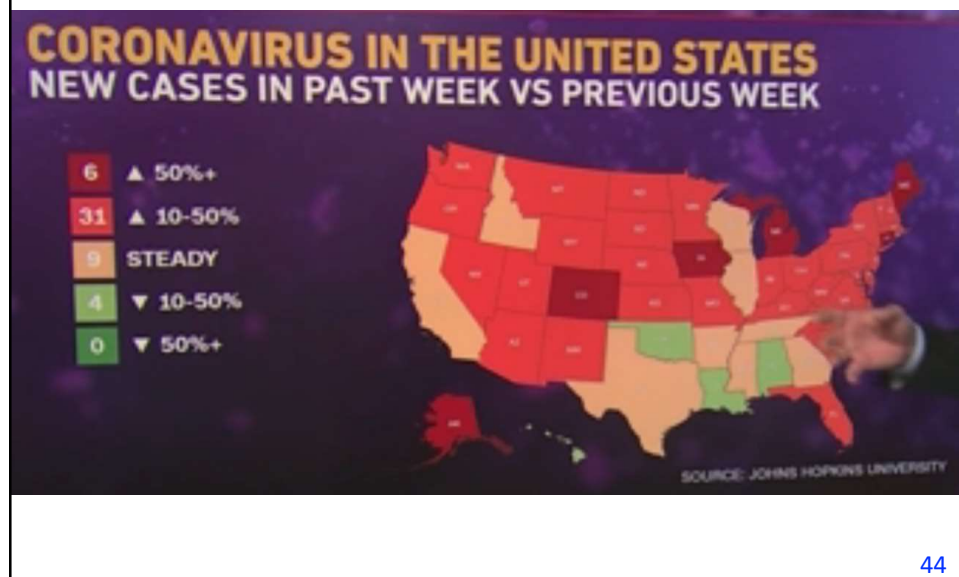
- District of Columbia
- New Hampshire
- Vermont
- Hawaii
- New York
- Washington
- Maine

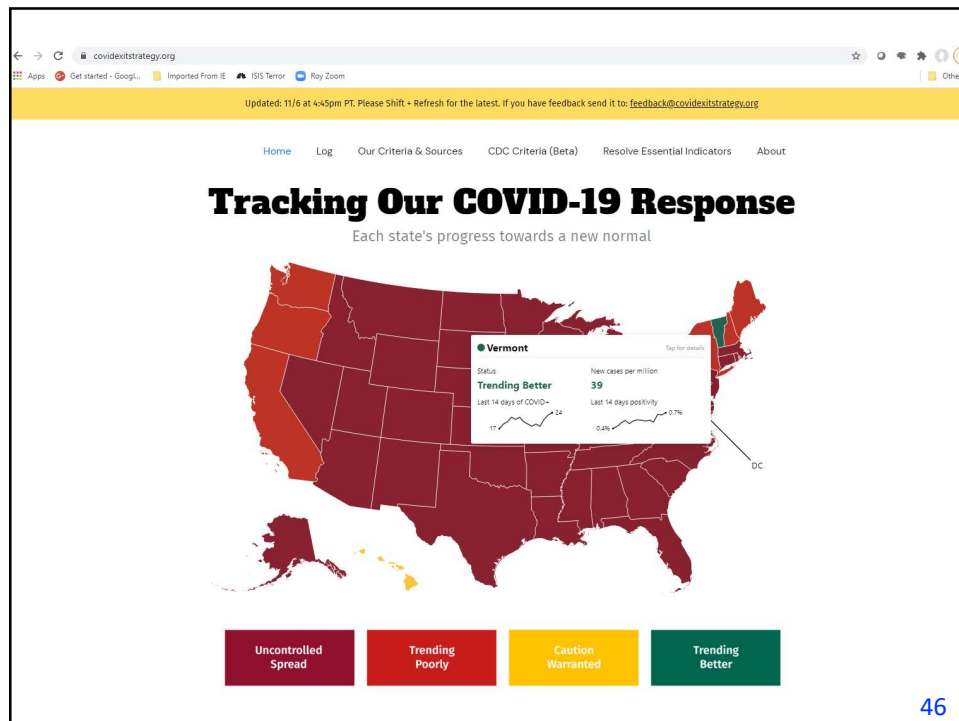
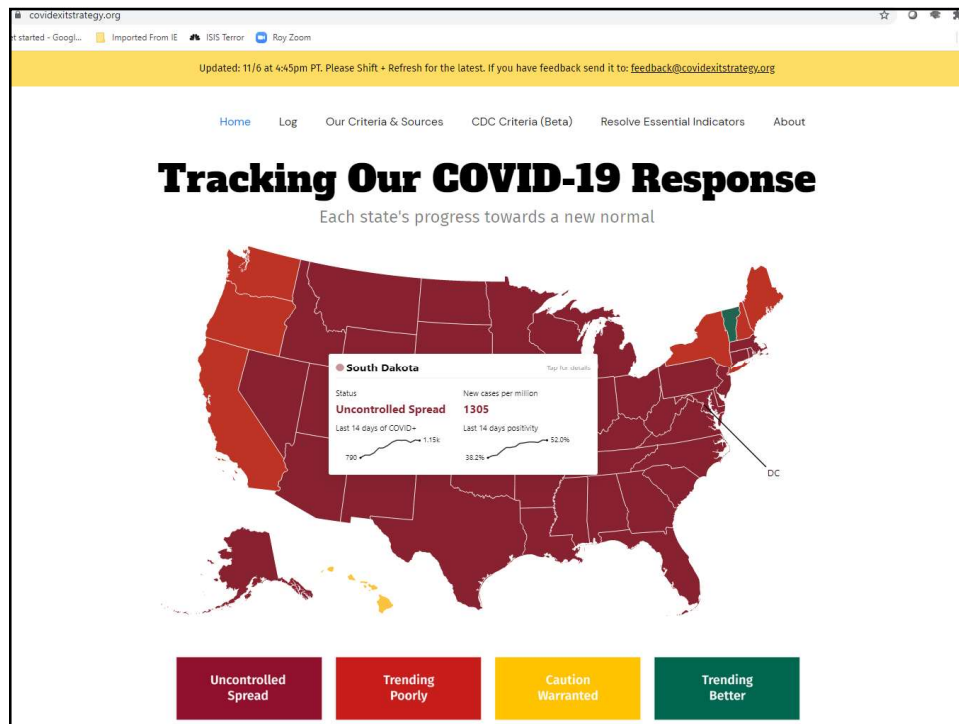
| County risk level                                                                                                       | New cases                                        | Positive tests                        |
|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|---------------------------------------|
| <b>WIDESPREAD</b><br>Most schools are closed to in-person instruction, unless they meet certain conditions              | <b>More than 7</b><br>daily new cases (per 100k) | <b>More than 8%</b><br>positive tests |
| <b>SUBSTANTIAL</b><br>Schools can reopen for in-person instruction after they have remained in the red tier for 14 days | <b>4 - 7</b><br>daily new cases (per 100k)       | <b>5 - 8%</b><br>positive tests       |
| <b>MODERATE</b><br>Schools can reopen for in-person instruction based on state and county guidance                      | <b>1 - 3.9</b><br>daily new cases (per 100k)     | <b>2 - 4.9%</b><br>positive tests     |
| <b>MINIMAL</b><br>Schools can reopen for in-person instruction based on state and county guidance                       | <b>Less than 1</b><br>daily new cases (per 100k) | <b>Less than 2%</b><br>positive tests |

Source: California Department of Public Health (Covid19.CA.GOV)



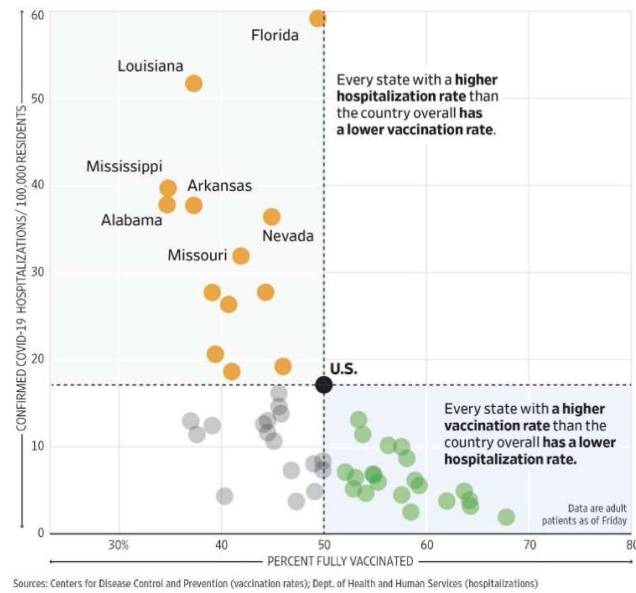
CNN: What's bad about this?





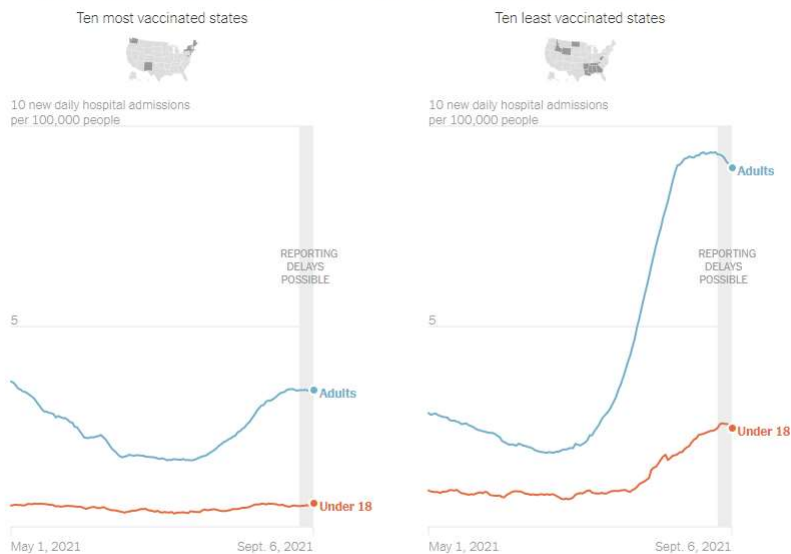


## Vaccination and Hospitalization



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### Covid hospital admissions for children are climbing in states with low immunization rates



Sources: U.S. Department of Health and Human Services; U.S. Census Bureau - Seven-day averages.  
Reporting delays may affect data from the past seven days. States are ordered by the share of all residents who are fully vaccinated. Data includes hospital admissions of those with either confirmed or suspected Covid-19.



# Data Visualization

- Definition: The visual representation and presentation of data to facilitate understanding
- Data Mining
  - Predictive Analytics
  - Descriptive Analytics
- Context for Data Visualization
- Why Visualization?
- Preattentive Attributes
- Use of Color
- Maps
- Types of Charts
- Potential for Misperception
- Examples
- Exercise
- Material
  - The Big Book of Dashboards, by Steve Wexler et. al.
  - Storytelling with Data, by Cole Nussbaumer Knaflic

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## Why do we need to Visualize Data?

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## Context for Data Visualization

- Data visualization is the last step of a larger process known as **knowledge discovery in databases (KDD)**.
- The KDD process model comprises six phases
  - Data selection
  - Data cleansing
  - Enrichment
  - Data transformation or encoding
  - Data mining
  - **Reporting and displaying discovered knowledge**

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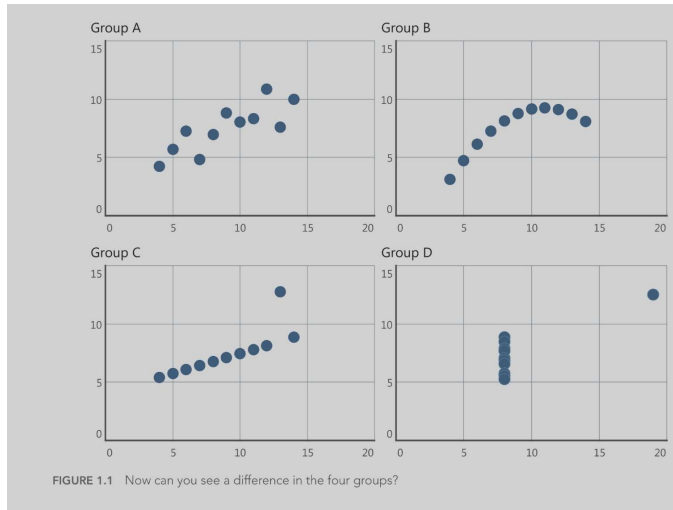
## What do these numbers tell you?

TABLE 1.1 Table with four groups of numbers: What do they tell you?

| Group A |       | Group B |      | Group C |       | Group D |       |
|---------|-------|---------|------|---------|-------|---------|-------|
| x       | y     | x       | y    | x       | y     | x       | y     |
| 10.00   | 8.04  | 10.00   | 9.14 | 10.00   | 7.46  | 8.00    | 6.58  |
| 8.00    | 6.95  | 8.00    | 8.14 | 8.00    | 6.77  | 8.00    | 5.76  |
| 13.00   | 7.58  | 13.00   | 8.74 | 13.00   | 12.74 | 8.00    | 7.71  |
| 9.00    | 8.81  | 9.00    | 8.77 | 9.00    | 7.11  | 8.00    | 8.84  |
| 11.00   | 8.33  | 11.00   | 9.26 | 11.00   | 7.81  | 8.00    | 8.47  |
| 14.00   | 9.96  | 14.00   | 8.10 | 14.00   | 8.84  | 8.00    | 7.04  |
| 6.00    | 7.24  | 6.00    | 6.13 | 6.00    | 6.08  | 8.00    | 5.25  |
| 4.00    | 4.26  | 4.00    | 3.10 | 4.00    | 5.39  | 19.00   | 12.50 |
| 12.00   | 10.84 | 12.00   | 9.13 | 12.00   | 8.15  | 8.00    | 5.56  |
| 7.00    | 4.82  | 7.00    | 7.26 | 7.00    | 6.42  | 8.00    | 7.91  |
| 5.00    | 5.68  | 5.00    | 4.74 | 5.00    | 5.73  | 8.00    | 6.89  |

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## Anscombe's Quartet



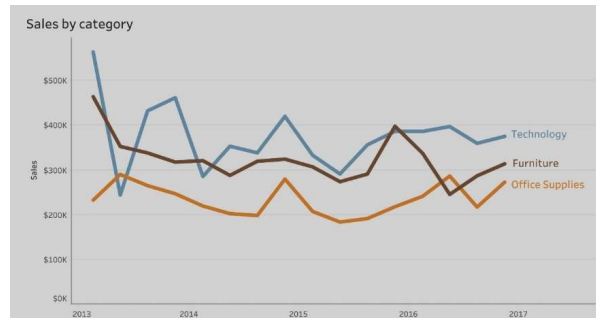
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## What are the Trends in Sales?

| Category        | 2013 Q1   | 2013 Q2   | 2013 Q3   | 2013 Q4   | 2014 Q1   | 2014 Q2   | 2014 Q3   | 2014 Q4   |
|-----------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| Furniture       | \$463,988 | \$352,779 | \$338,169 | \$317,735 | \$320,875 | \$287,934 | \$319,537 | \$324,319 |
| Office Supplies | \$232,558 | \$290,055 | \$265,083 | \$246,946 | \$219,514 | \$202,412 | \$198,268 | \$279,679 |
| Technology      | \$563,866 | \$244,045 | \$432,299 | \$461,616 | \$285,527 | \$353,237 | \$338,360 | \$420,018 |
| Category        | 2015 Q1   | 2015 Q2   | 2015 Q3   | 2015 Q4   | 2016 Q1   | 2016 Q2   | 2016 Q3   | 2016 Q4   |
| Furniture       | \$307,028 | \$273,836 | \$290,886 | \$397,912 | \$337,299 | \$245,445 | \$286,972 | \$313,878 |
| Office Supplies | \$207,363 | \$183,631 | \$191,405 | \$217,950 | \$241,281 | \$286,548 | \$217,198 | \$272,870 |
| Technology      | \$333,002 | \$291,116 | \$356,243 | \$386,445 | \$386,387 | \$397,201 | \$359,656 | \$375,229 |

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## Graphed Sales Data



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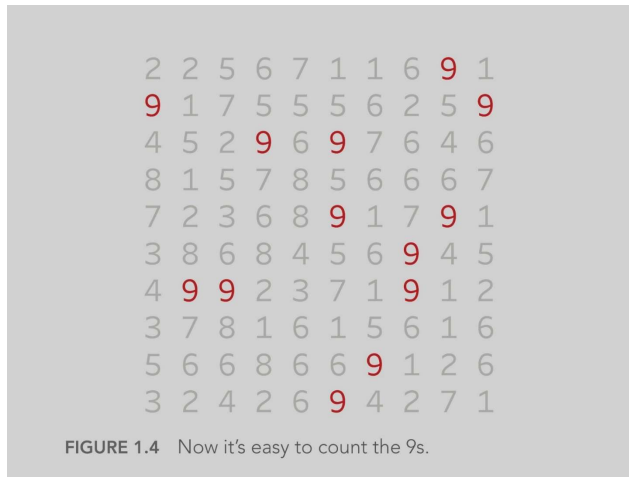
Poll:  
How many 9's are there?

|   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 2 | 5 | 6 | 7 | 1 | 1 | 6 | 9 | 1 |
| 9 | 1 | 7 | 5 | 5 | 5 | 6 | 2 | 5 | 9 |
| 4 | 5 | 2 | 9 | 6 | 9 | 7 | 6 | 4 | 6 |
| 8 | 1 | 5 | 7 | 8 | 5 | 6 | 6 | 6 | 7 |
| 7 | 2 | 3 | 6 | 8 | 9 | 1 | 7 | 9 | 1 |
| 3 | 8 | 6 | 8 | 4 | 5 | 6 | 9 | 4 | 5 |
| 4 | 9 | 9 | 2 | 3 | 7 | 1 | 9 | 1 | 2 |
| 3 | 7 | 8 | 1 | 6 | 1 | 5 | 6 | 1 | 6 |
| 5 | 6 | 6 | 8 | 6 | 6 | 9 | 1 | 2 | 6 |
| 3 | 2 | 4 | 2 | 6 | 9 | 4 | 2 | 7 | 1 |

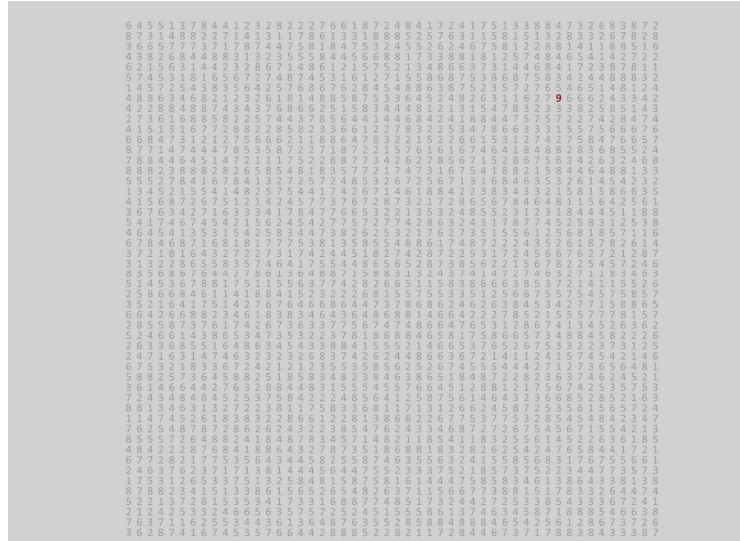
56



## How many 9's now?



# This time?



59

# Size works too

|   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 2 | 5 | 6 | 7 | 1 | 1 | 6 | 9 | 1 |
| 9 | 1 | 7 | 5 | 5 | 5 | 6 | 2 | 5 | 9 |
| 4 | 5 | 2 | 9 | 6 | 9 | 7 | 6 | 4 | 6 |
| 8 | 1 | 5 | 7 | 8 | 5 | 6 | 6 | 6 | 7 |
| 7 | 2 | 3 | 6 | 8 | 9 | 1 | 7 | 9 | 1 |
| 3 | 8 | 6 | 8 | 4 | 5 | 6 | 9 | 4 | 5 |
| 4 | 9 | 9 | 2 | 3 | 7 | 1 | 9 | 1 | 2 |
| 3 | 7 | 8 | 1 | 6 | 1 | 5 | 6 | 1 | 6 |
| 5 | 6 | 6 | 8 | 6 | 6 | 9 | 1 | 2 | 6 |
| 3 | 2 | 4 | 2 | 6 | 9 | 4 | 2 | 7 | 1 |

60

## What's going on?

- Size and hue are preattentive attributes
- Preattentive attributes just “pop out” at you
- How are preattentive attributes processed differently?

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## Q in O's

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| O | O | O | O | O | O | O |
| O | O | O | O | O | O | O |
| O | O | O | O | O | O | O |
| O | O | O | O | O | O | O |
| O | O | O | O | Q | O | O |
| O | O | O | O | O | O | O |
| O | O | O | O | O | O | O |

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## O in Q's

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| Q | Q | Q | Q | Q | Q | Q |
| Q | Q | Q | Q | Q | Q | Q |
| Q | Q | Q | Q | Q | O | Q |
| Q | Q | Q | Q | Q | Q | Q |
| Q | Q | Q | Q | Q | Q | Q |
| Q | Q | Q | Q | Q | Q | Q |
| Q | Q | Q | Q | Q | Q | Q |

63

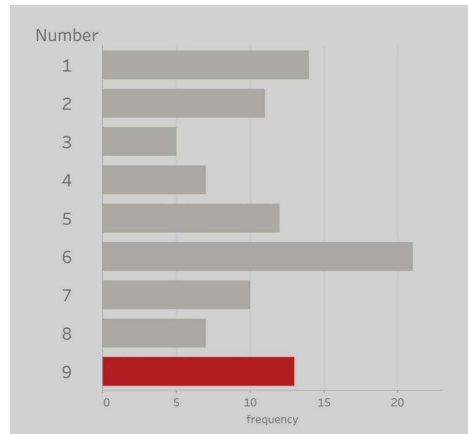
## Can Overdue It

|   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 2 | 5 | 6 | 7 | 1 | 1 | 6 | 9 | 1 |
| 9 | 1 | 7 | 5 | 5 | 5 | 6 | 2 | 5 | 9 |
| 4 | 5 | 2 | 9 | 6 | 9 | 7 | 6 | 4 | 6 |
| 8 | 1 | 5 | 7 | 8 | 5 | 6 | 6 | 6 | 7 |
| 7 | 2 | 3 | 6 | 8 | 9 | 1 | 7 | 9 | 1 |
| 3 | 8 | 6 | 8 | 4 | 5 | 6 | 9 | 4 | 5 |
| 4 | 9 | 9 | 2 | 3 | 7 | 1 | 9 | 1 | 2 |
| 3 | 7 | 8 | 1 | 6 | 1 | 5 | 6 | 1 | 6 |
| 5 | 6 | 6 | 8 | 6 | 6 | 9 | 1 | 2 | 6 |
| 3 | 2 | 4 | 2 | 6 | 9 | 4 | 2 | 7 | 1 |

64

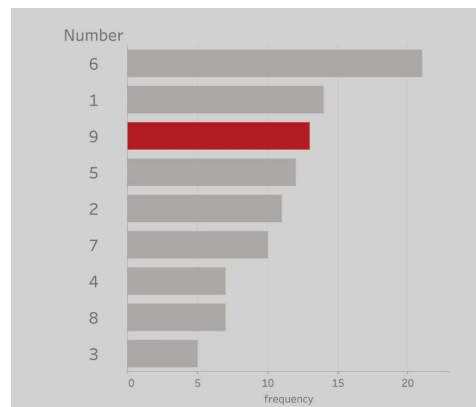


## Bar Chart

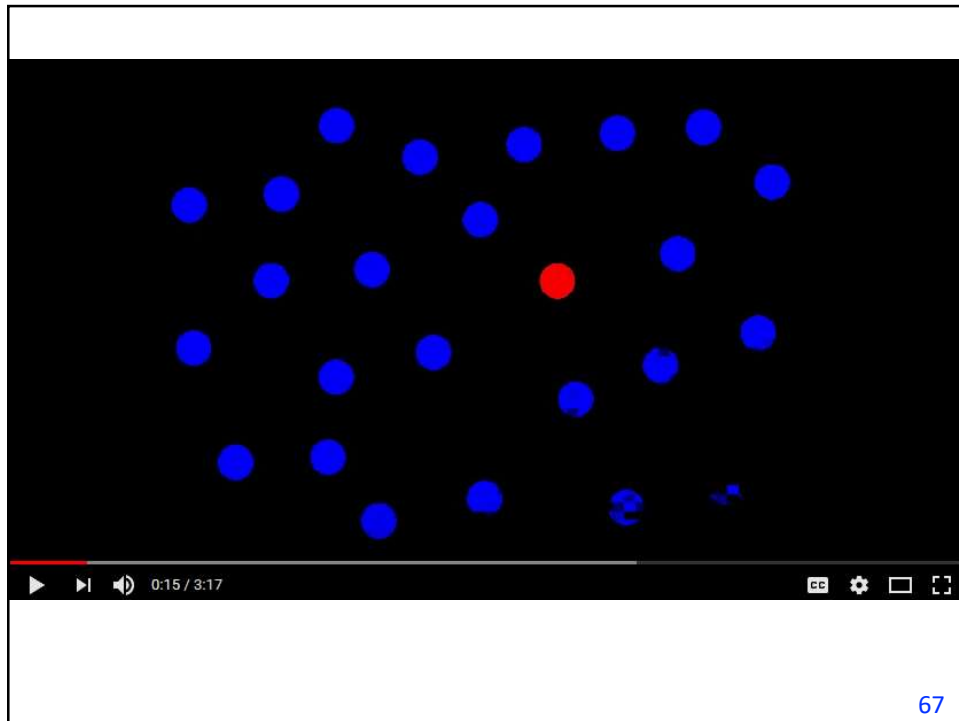


65

## Sorted Bar Chart



66

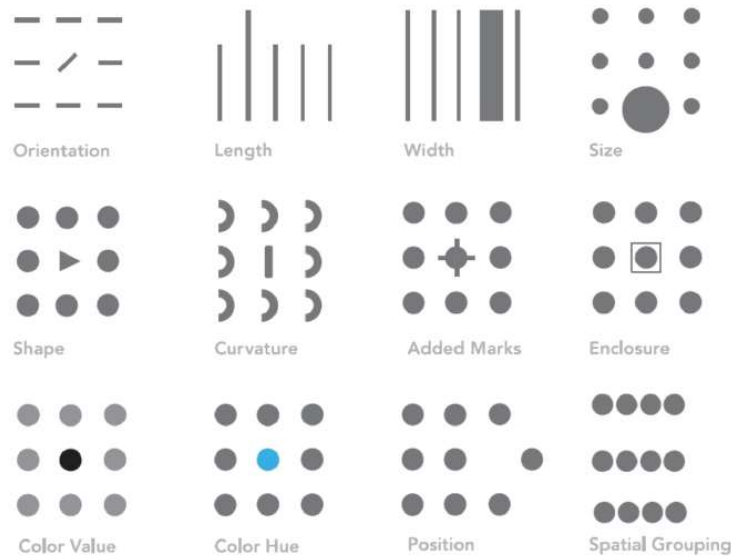


67



68

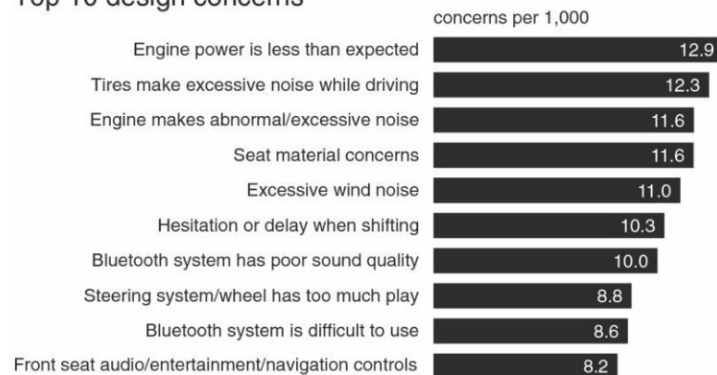
## Preattentive Attributes



69

## No Clear Message

### Top 10 design concerns



This is like counting 9s, i.e., nothing grabs your attention.

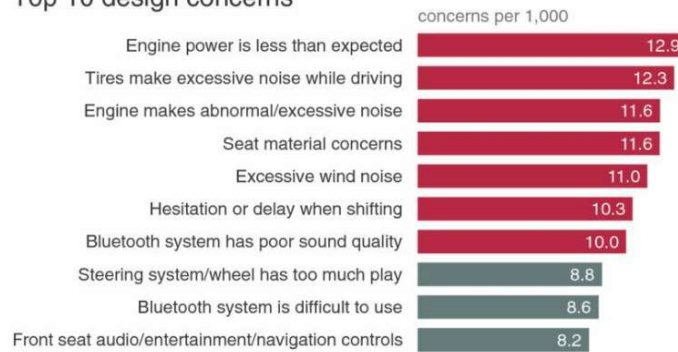
70

## More Use of Preattentive Attributes

7 of the top 10 design concerns have 10 or more concerns per 1,000.

Discussion: is this an acceptable default rate?

### Top 10 design concerns



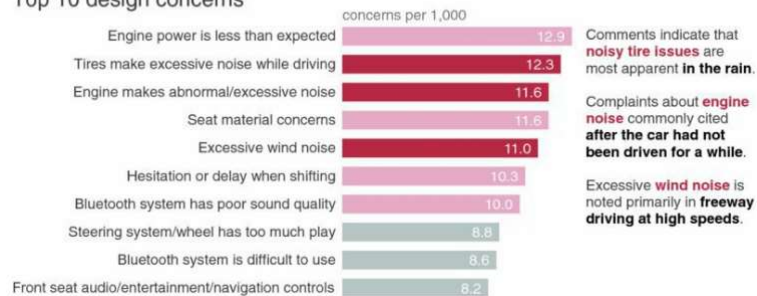
Note the tie between the color in the header and graph.

71

## Even More Use of Preattentive Attributes

Of the top design concerns, three are noise-related.

### Top 10 design concerns

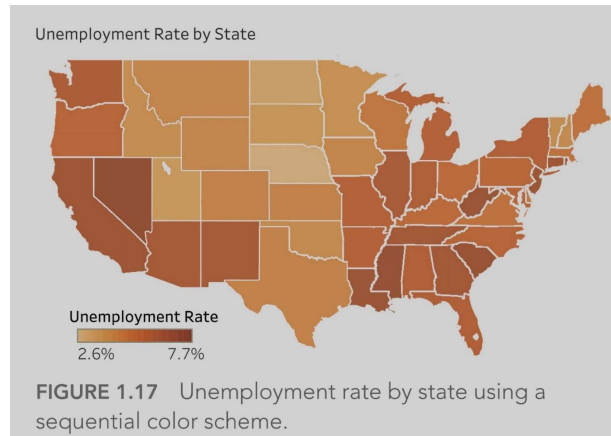


Here color and bold are used to deliver a different message.

72

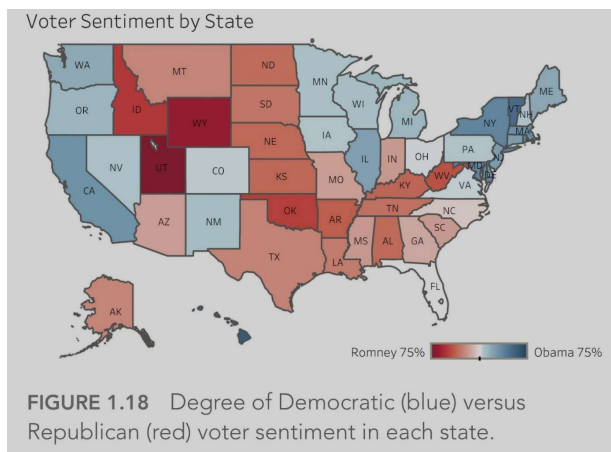


## Sequential Use of Color in Maps



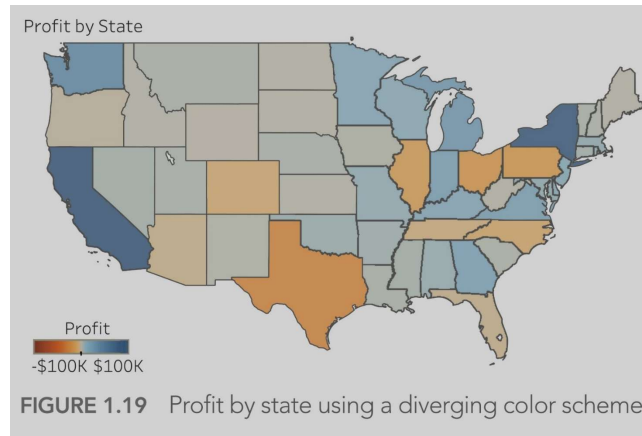
75

## Diverging Use of Color in Maps I



76

## Diverging Use of Color in Maps II

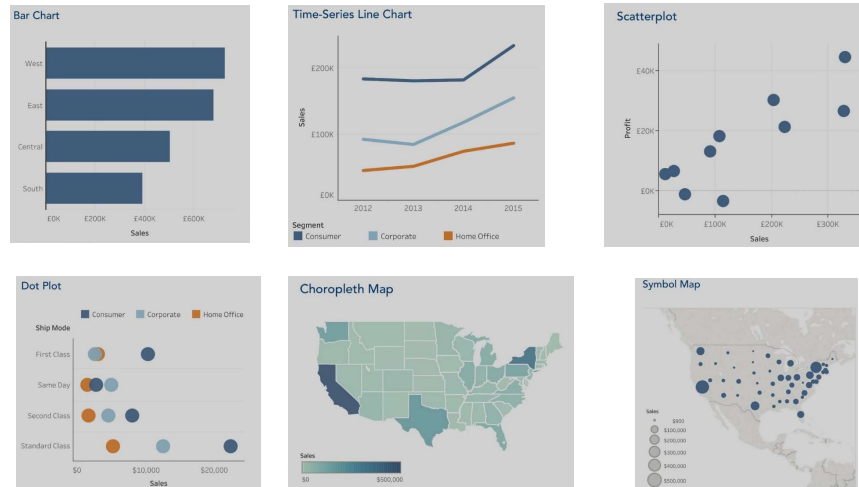


77



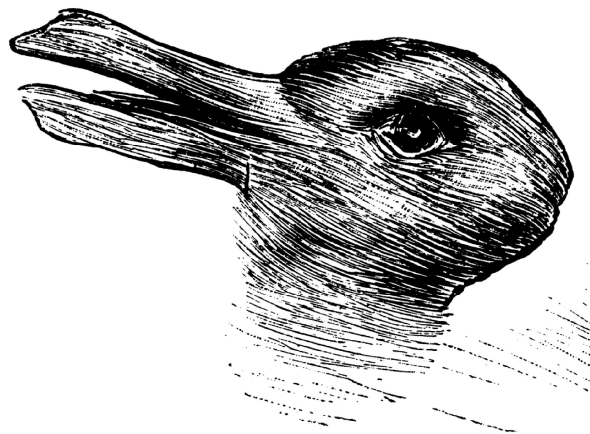
78

## Common Chart Types



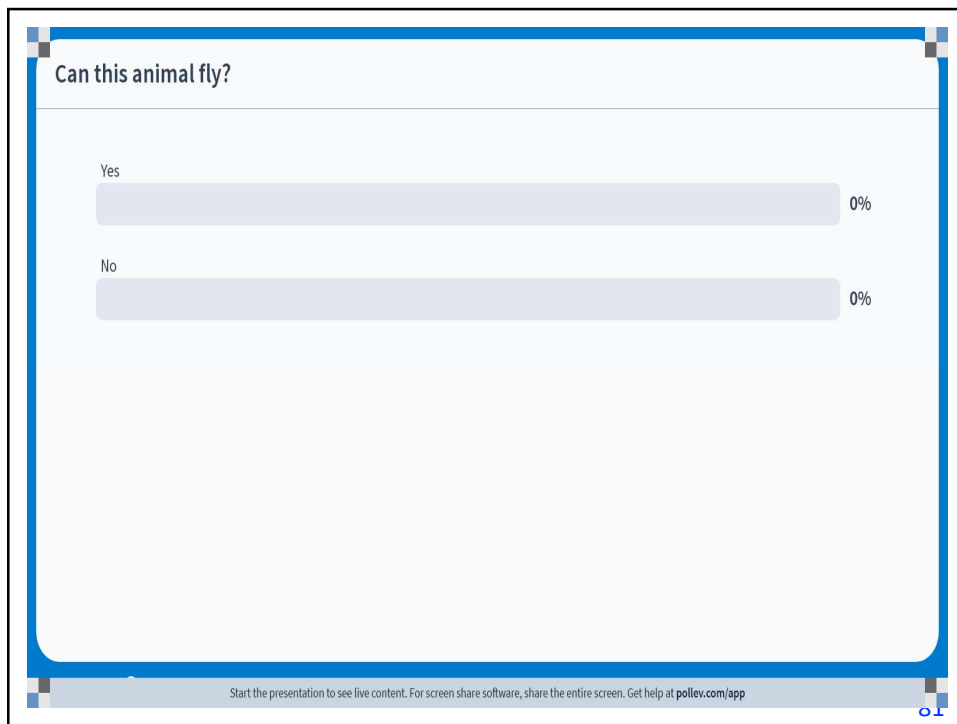
79

Can this animal fly?

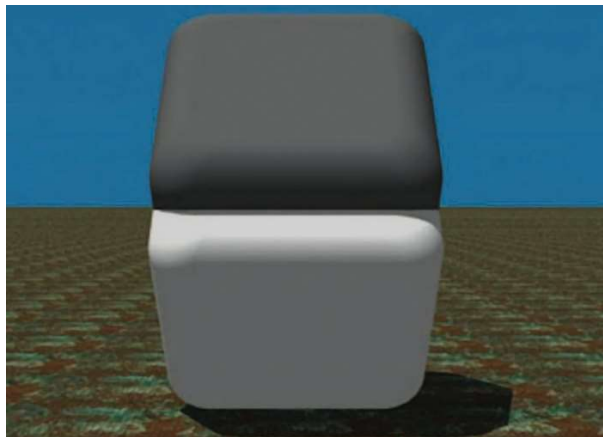


80

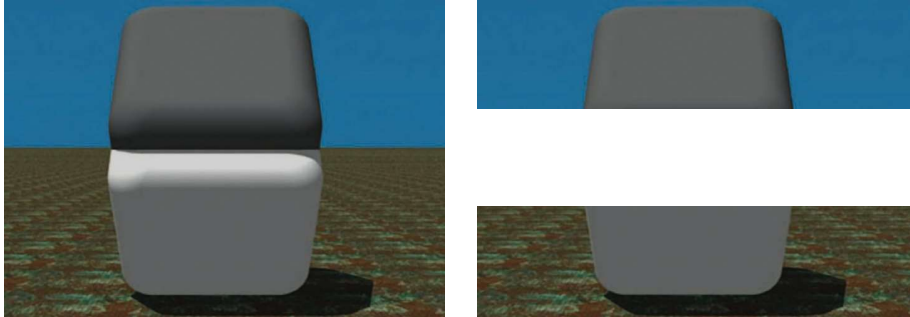




## Mind Can Play Tricks



## Proof



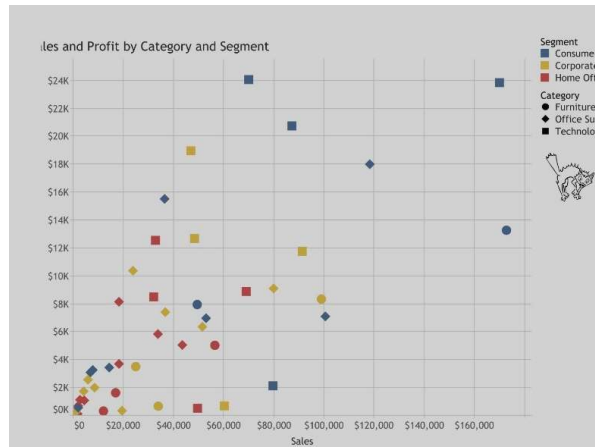
83

## You See What You Expect to See



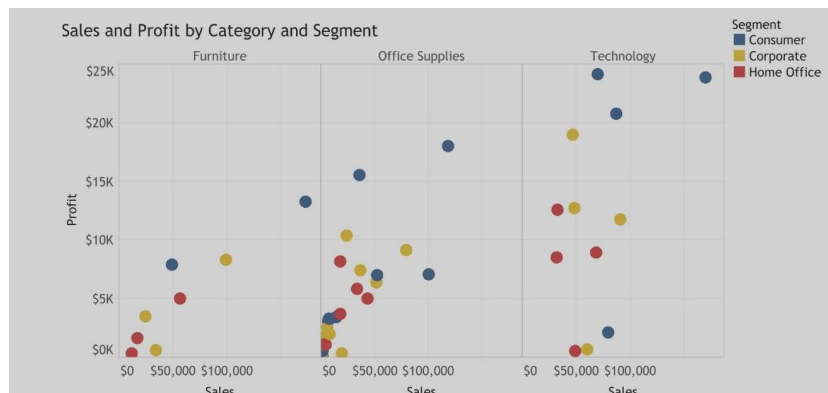
84

## Sales and Profit by Category and Segment



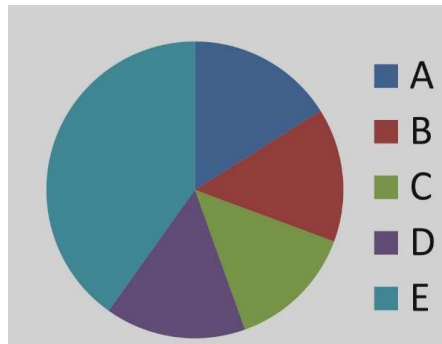
85

## Improved Sales and Profit



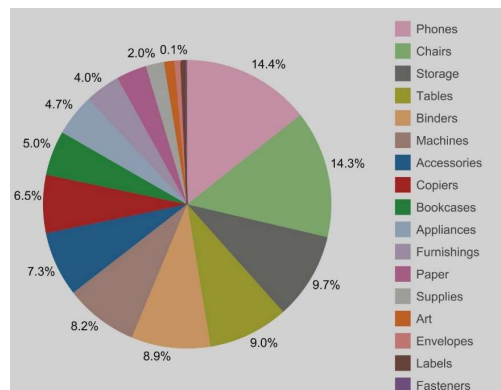
86

## Pie Chart I



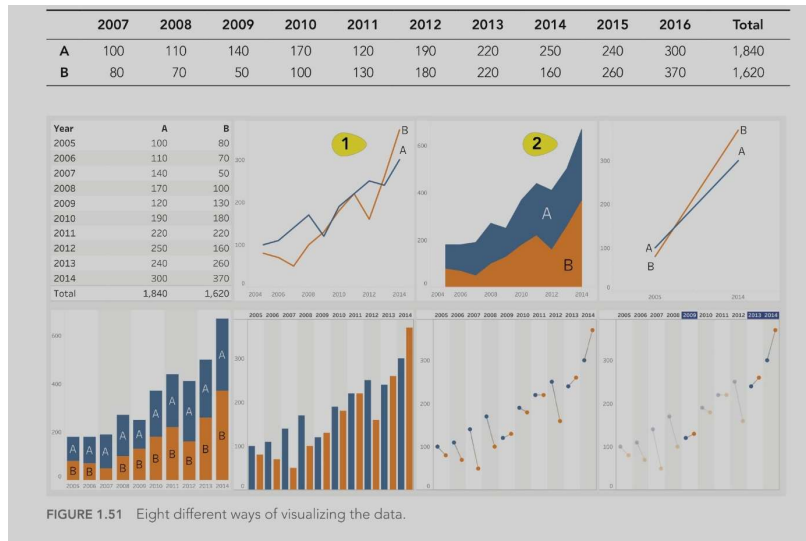
87

## How often do you see something like this?



88

## Visualization Tradeoffs: Pros and Cons of 1 and 2?



89

**TED** Ideas worth spreading

WATCH DISCOVER

Share

Add to list

Like

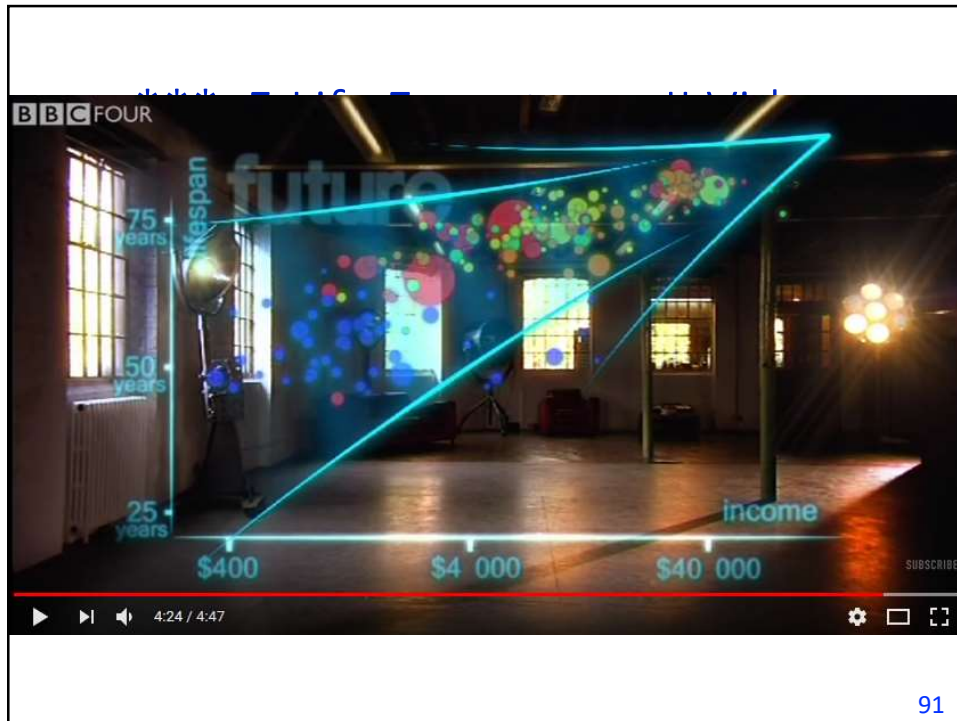
Rate

Hans Rosling at TED2006

**The best stats you've ever seen**

19:44

90



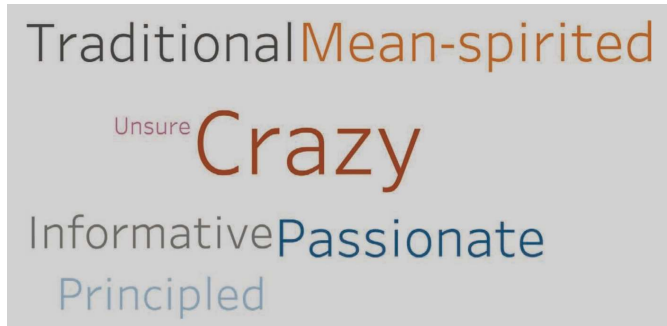
91

## Poll on Views of Election

|               |     |
|---------------|-----|
| Crazy         | 40% |
| Mean-spirited | 14% |
| Passionate    | 13% |
| Traditional   | 13% |
| Informative   | 9%  |
| Principled    | 9%  |
| Unsure        | 2%  |

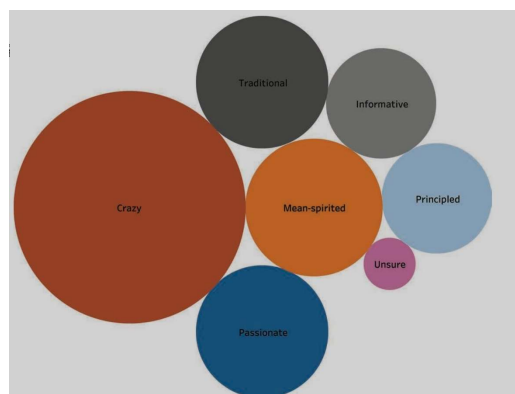
92

## Word Cloud



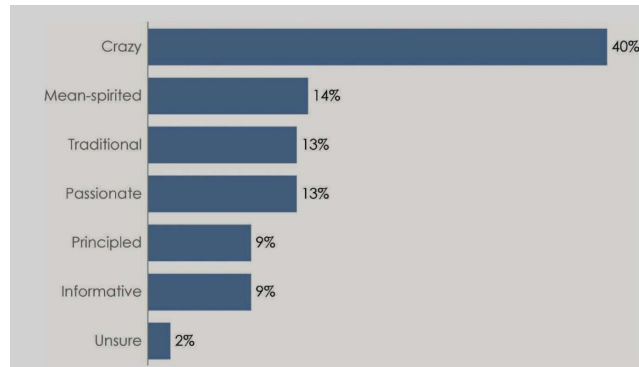
93

## Bubble Charts



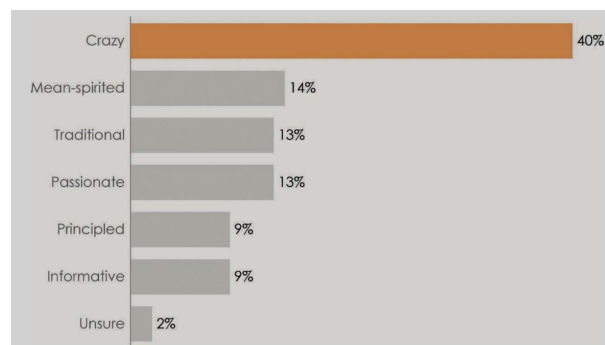
94

## Simple Bar Chart



95

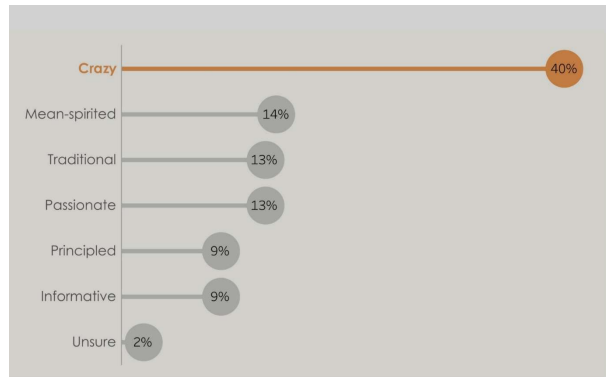
## Colored Bar Chart



96

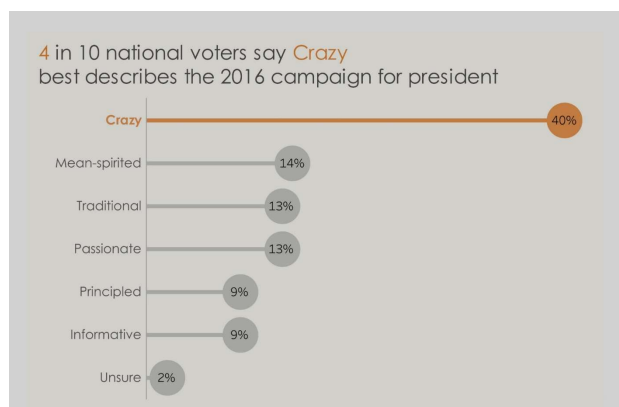


## Lollipop Chart



97

## Lollipop Chart with Descriptive Headline



98

## Exercise: What's Wrong With This?

|      |          |         |         |
|------|----------|---------|---------|
| East | Region C | 54.1%   | ▲ 11.1  |
|      | Region E | 47.3%   | ▼ -2.0  |
|      | Region D | 31.7%   | ▲ 3.8   |
|      | Region B | 29.6%   | ▼ -0.5  |
|      | Region A | 26.0%   | ▼ -1.1  |
| West | Region F | 47.8%   | ▲ 7.4   |
|      | Region M | 47.3%   | ▲ 1.3   |
|      | Region L | 47.1%   | ▲ 4.2   |
|      | Region G | 44.3%   | ▲ 0.4   |
|      | Region K | 42.9%   | ▼ -1.2  |
|      | Region N | 42.6%   | ▼ -10.8 |
|      | Region J | 42.3%   | ▲ 4.2   |
|      | Region H | 40.3%   | ▼ -4.6  |
|      |          | Current | QoQ     |

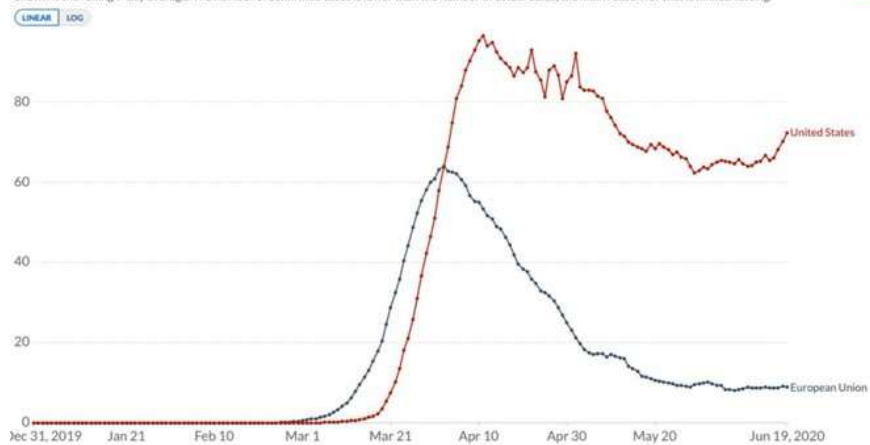
99

## US vs Europe through June 2020

Daily new confirmed COVID-19 cases per million people

Shown is the rolling 7-day average. The number of confirmed cases is lower than the number of actual cases; the main reason for that is limited testing.

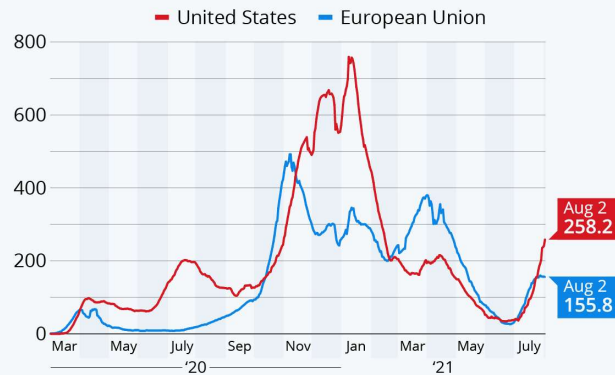
Our World  
in Data



100

## U.S. Cases Keep Rising, Surpassing Last Summer's Peak

Seven-day rolling average of newly confirmed COVID-19 cases per million people in the U.S. and the EU



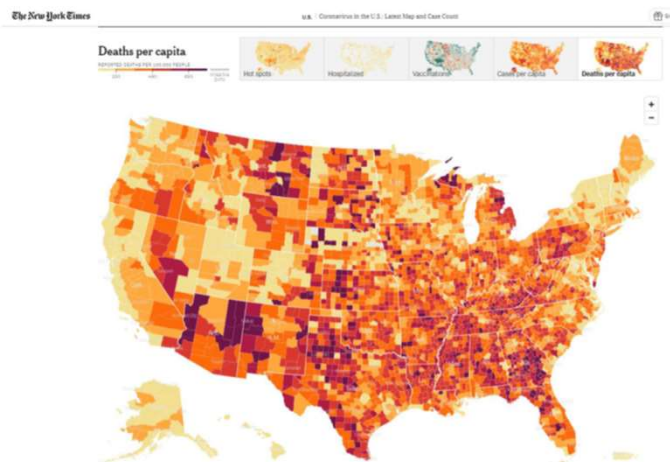
Source: Our World in Data



statista

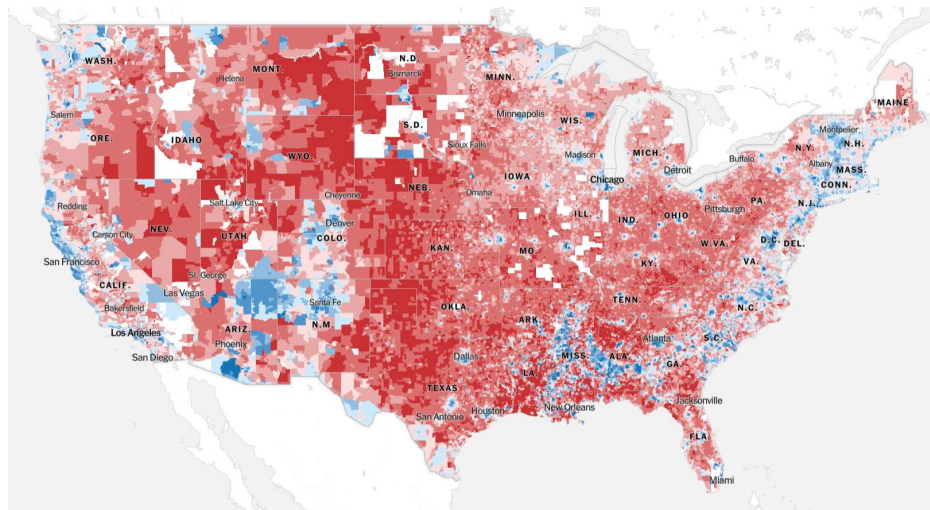
101

## Deaths per capita



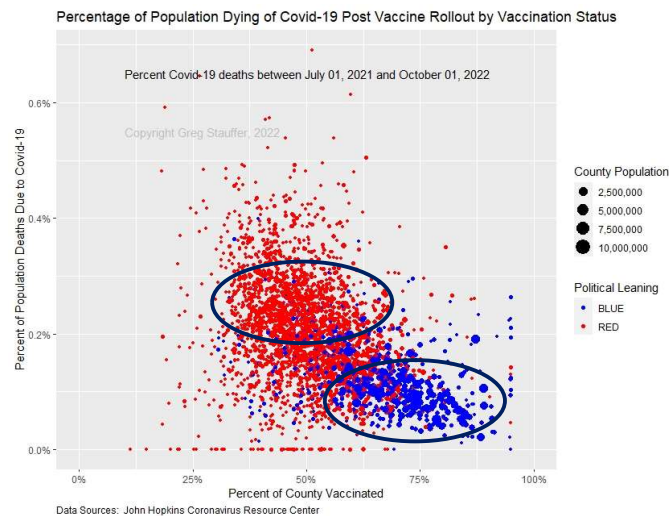
102

## 2024 Presidential Voting



103

## Deaths per capita vs. Percent of County Vaccinated and Political Leaning



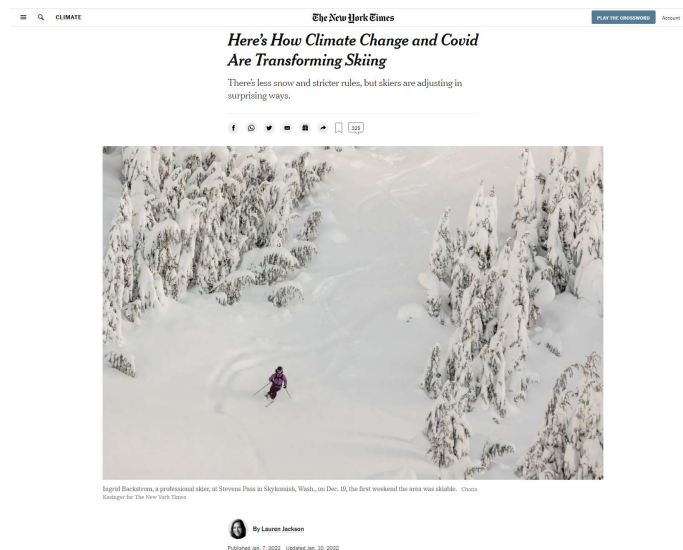
104

## Principles of Data Visualization

1. Understand the context
2. Choose an appropriate visual display
3. Eliminate clutter
4. Focus attention where you want it
5. Think like a designer
6. Tell a story

From: Storytelling with Data

105



Between 1982 and 2016, the American ski season shrunk by an average of 34 days, and levels of snow cover saw an average drop of 41 percent, according to [a study in the journal Geophysical Research Letters](#).

106

## Geophysical Research Letters

Research Letter | [Free Access](#)

### Snowpack Change From 1982 to 2016 Over Conterminous United States

Xubin Zeng  Patrick Broxton, Nicholas Dawson

First published: 12 December 2018 | <https://doi.org/10.1029/2018GL079621> | Citations: 31

[Find text@JGR](#)

 SECTIONS

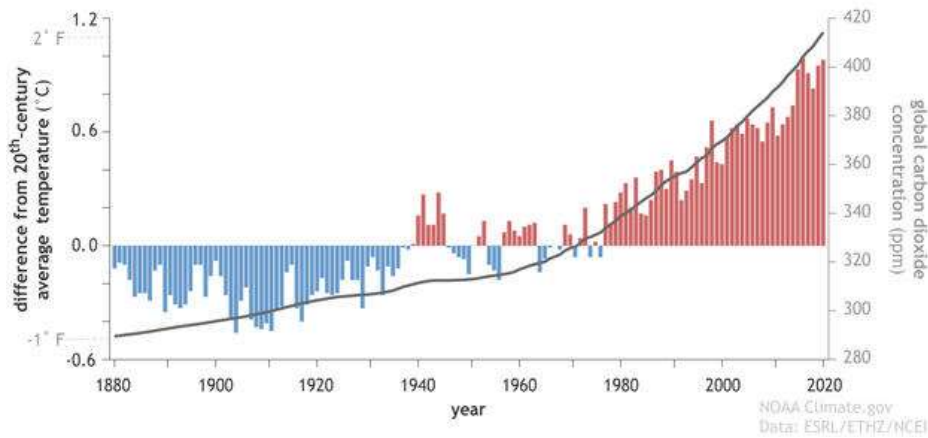
 PDF  TOOLS  SHARE

#### Abstract

Snow water equivalent (SWE) variability and its drivers over different regions remain uncertain due to lack of representativeness of point measurements and deficiencies of existing coarse-resolution SWE products. Here, for the first time, we quantify and understand the snowpack change from 1982 to 2016 over conterminous United States at 4-km pixels. Annual maximum SWE decreased significantly ( $p < 0.05$ ) by 41% on average for 13% of snowy pixels over western United States. Snow season was shortened significantly by 34 days on average for 9% of snowy pixels over the United States, primarily caused by earlier ending and later arrival of the season over western and eastern United States, respectively. October–March mean temperature and accumulated precipitation largely explain the temporal variability of 1 April SWE over western United States, and considering temperature alone would exaggerate the warming effect on SWE decrease. In contrast, temperature plays the primary role in the 1 April SWE variability over eastern United States.

107

Global atmospheric carbon dioxide and surface temperature (1880–2020)



108

## Wrap-up

- Data visualization is important
- Analysis is only as good as your ability to sell it
- Short assignment to watch and make a brief comment on five videos will be due next Tuesday night
  - Will be available sometime Friday
  - Should take under an hour
- Next Tuesday – Start working in Tableau
  - Will send out an email how to get access to Tableau no later than Monday
- Next Thursday – AI Project discussion
- Questions?